

Lesson 2.3: Determinants

Learning Outcomes: At the end of the lesson, the students are able to:

- 1. evaluate determinants using the permutation definition;
- 2. interpret determinants geometrically.
- 3. establish properties of determinants.

Permutations

Informally speaking, a **permutation** on the finite set $X = \{1, 2, \dots, n\}$ is an *arrangement* of all the elements X in a row. For example, consider the set

$$X = \{1, 2, 3\}.$$

The set X has six distinct permutations represented as follows:

123	213	312
132	321	213

To define the *determinant* of a square matrix using permutations, we would like to think of a permutation, say $\rho = 231$ as a mapping or a function $\rho : X \longrightarrow X$ such that $\rho(1) = 2$, $\rho(2) = 3$, and $\rho(3) = 1$. We can imagine ρ as assignment of 3 applicants to 3 vacant job positions such that no two applicants land on the same job. With this in mind, we have the following operational definition.

Definition 103: Permutation on a Finite Set

A **permutation** on a set $X = \{1, 2, \dots, n\}$ is any bijective function $\pi : X \longrightarrow X$. The set of all permutations on X is denoted by S_n .

Note that since X is finite, any one-to-one function from X to X is already a bijective function. If $\pi \in S_n$, we shall write

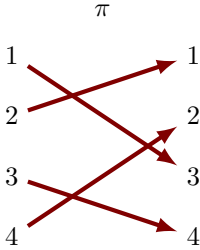
$$\pi = \pi(1)\pi(2) \cdots \pi(n).$$

Example 104.

The permutation $\pi = 3142 \in S_4$ is the function

$$\pi : \{1, 2, 3, 4\} \longrightarrow \{1, 2, 3, 4\}$$

such that $\pi(1) = 3$, $\pi(2) = 1$, $\pi(3) = 4$, and $\pi(4) = 2$. Using arrow diagrams, we could represent π as follows:



Think of this as if we have assigned applicant number 1 to position 3, applicant 2 to position 1, and so on.



Imagine constructing an arbitrary permutation $\pi \in S_n$. If all the applicants are yet to be assigned on any job position, applicant 1 may be assigned to n possible positions. If applicant 1 has been assigned, applicant 2 can only be assigned to $n - 1$ positions. If the first two applicants have been assigned, applicant 3 can only be assigned to $n - 2$ positions, and so on. Thus, by the fundamental counting principle, there are a total of

$$n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 2 \cdot 1 = n!$$

possible ways to assign n applicants to n job positions, given that no two applicants are assigned to the same position. As a remark, there are exactly $n!$ permutations on the set $X = \{1, 2, \dots, n\}$, i.e.

$$|S_n| = n!.$$

In real life, job assignments are not done arbitrarily. Let us now make sense of the order in the set $\{1, 2, \dots, n\}$. As applicants, the higher the value would mean a higher qualification and as job positions, the higher the value would mean a job paying higher salary. In an ideal world, where injustice is not present, the natural scenario would be to assign applicant i to job position i , for $i = 1, 2, \dots, n$.

Consider a job assignment π such that there exists a pair of applicants (i, j) , where $i < j$, but $\pi(i) > \pi(j)$. This would mean that applicant i was assigned to a job paying more, even though applicant j has a higher qualification than applicant i . **This is injustice in all levels**, and we should not allow this as decent human beings! However, we don't have control to all the things in the world. We can only give a name to it.

Definition 105: Inversions

Let $\pi \in S_n$. We say that π performs an **inversion**, if there exists an ordered pair $(i, j) \in \{1, 2, \dots, n\} \times \{1, 2, \dots, n\}$ such that $i < j$, but $\pi(i) > \pi(j)$.

Example 106. Let $\pi = 15324 \in S_5$. π performs the following inversions:

- $(2, 3)$: $2 < 3$, but $\pi(2) = 5 > 3 = \pi(3)$.
- $(2, 4)$: $2 < 4$, but $\pi(2) = 5 > 2 = \pi(4)$.
- $(2, 5)$: $2 < 5$, but $\pi(2) = 5 > 4 = \pi(5)$.
- $(3, 4)$: $3 < 4$, but $\pi(3) = 3 > 2 = \pi(4)$.

We would like to classify permutations according to the number of inversions it performs.

Definition 107

A permutation $\pi \in S_n$ is **even** if it performs an even number of inversions. Otherwise, π is **odd**.

Example 108. The permutation $\pi = 15324 \in S_5$ is even since it performs exactly 4 inversions. On the other hand, the permutation $\sigma = 213 \in S_3$ is odd since it performs exactly 1 inversion.

Example 109. For every $n > 1$, the **identity permutation** $\iota = 123 \cdots n \in S_n$ is even since it performs 0 inversion (which is even).

The following are facts derived from an elementary course in group theory. We shall not prove them here.

Theorem 110

1. For each $n > 1$, the set of even permutations in S_n and the set of odd permutations in S_n have the same number of elements.
2. If two elements in an even (odd) permutation are switched, then the resulting permutation is odd (even, resp.).

Determinant of an n -by- n Matrix

We are now ready to define the determinant of an n -by- n matrix.

Definition 111: Determinant

Let $A = [a_{ij}] \in M_n$. The **determinant** of A is defined as the scalar

$$\det(A) = |A| = \sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} a_{2,\pi(2)} \cdots a_{n,\pi(n)},$$

$$\text{where } (-1)^\pi := \begin{cases} 1 & \text{if } \pi \text{ is even;} \\ -1 & \text{if } \pi \text{ is odd.} \end{cases}$$

With our knowledge of the properties of permutations, we can give the following as remarks.

1. The sum is taken over all permutations in S_n .
2. Every term of $\det(A)$ is a *signed* product of n entries obtained by taking one entry from each row such that no two entries are in the same column.
3. The expansion of $\det(A)$ has $n!$ terms in it. Half of the terms are preceded by a $+$ sign, and the other half are preceded by a $-$ sign.

Note that if $A = [a_{11}]$ is a 1-by-1 matrix, then $\det(A) = a_{11}$.

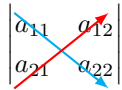
Consider the case when A is a 2-by-2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Since $n = 2$, we consider $S_2 = \{\textcolor{blue}{12}, \textcolor{red}{21}\}$. Note that the permutation $\textcolor{blue}{12}$ is even and $\textcolor{red}{21}$ is odd. Therefore,

$$\det(A) = (+1)a_{11}a_{22} + (-1)a_{12}a_{21} = a_{12}a_{22} - a_{12}a_{21}.$$

A good mnemonics to remember is to use the following criss-cross pattern of arrows.



$$= a_{11}a_{22} - a_{12}a_{21}.$$

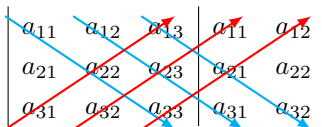
Let us now consider the case of a 3-by-3 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

Here $n = 3$, so we consider $S_3 = \{\textcolor{blue}{123}, \textcolor{blue}{231}, \textcolor{blue}{312}, \textcolor{red}{321}, \textcolor{red}{132}, \textcolor{red}{213}\}$. The blue-colored permutations are the even permutations, while the red ones are the odd permutations. Thus by definition,

$$\begin{aligned} \det(A) &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ &\quad - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}. \end{aligned}$$

We can construct a good mnemonics just like in the 2-by-2 case by repeating the first two columns of A and using the following criss-cross pattern of arrows.



$$\begin{aligned} &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ &\quad - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}. \end{aligned}$$

The blue arrows correspond to terms preceded by a $(+)$ sign, while the red arrows correspond to terms preceded by a $(-)$ sign.

Let us now implement these formulas through examples.

Example 112. Calculate the following determinants.

1. $\begin{vmatrix} 4 & -2 \\ 5 & -3 \end{vmatrix}$

3. $\begin{vmatrix} -5 & 1 & 2 \\ 1 & 2 & 3 \\ -4 & 3 & 5 \end{vmatrix}$

2. $\begin{vmatrix} 2 & -1 & 3 \\ 1 & 5 & 0 \\ -2 & 1 & 4 \end{vmatrix}$

4. $\begin{vmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & -3 & 0 & 0 \\ 2 & 3 & 4 & 5 \end{vmatrix}$

Solutions.

1. $\begin{vmatrix} 4 & -2 \\ 5 & -3 \end{vmatrix} = (4)(-3) - (5)(-2) = -12 + 10 = -2.$
2. For this, we repeat the first two columns and calculate the downward arrow terms and the upward arrow terms as follows:

downward ar- row terms	upward arrow terms
$(2)(5)(4) = 40$	$(-2)(5)(3) = -30$
$(-1)(0)(-2) = 0$	$(1)(0)(2) = 0$
$(3)(1)(1) = 3$	$(4)(1)(-1) = -4$

Therefore,

$$\begin{vmatrix} 2 & -1 & 3 \\ 1 & 5 & 0 \\ -2 & 1 & 4 \end{vmatrix} = (40 + 0 + 3) - (-30 + 0 + (-4)) = 77.$$

3. Using a similar procedure as in item (2), we get

$$\begin{vmatrix} -5 & 1 & 2 \\ 1 & 2 & 3 \\ -4 & 3 & 5 \end{vmatrix} = (-50 + (-12) + 6) - (-16 + (-45) + 5) = 0.$$

4.

$$\begin{vmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & -3 & 0 & 0 \\ 2 & 3 & 4 & 5 \end{vmatrix}$$

The matrix here is 4-by-4. Thus, any of the previous mnemonics will not work. We notice that the matrix has a lot of zeroes. In fact, exactly one element from each of rows 1 to 3 is non-zero.

By definition, every term of the determinant is a signed product of entries, one from each row and column. As we can see, most of the terms will be zero except for one the product

$$3 \cdot 4 \cdot (-3) \cdot 5.$$

Looking at the column numbers of the entries, we see that this term corresponds to the permutation $\pi = 1324$, which is odd. Therefore, the determinant has a value

$$(-1)^\pi \cdot 3 \cdot 4 \cdot (-3) \cdot 5 = (-1) \cdot 3 \cdot 4 \cdot (-3) \cdot 5 = 180.$$



Example 113. Let $A = [a_{ij}]$ be an upper triangular matrix, that is,

$$a_{ij} = 0 \text{ for } i < j$$

(0's above the main diagonal). Then $\det(A) = a_{11}a_{22} \cdots a_{nn}$ is just the product of the main diagonal entries because every term of $\det(A)$ of the form

$$(-1)^\pi a_{1,\pi(1)} a_{2,\pi(2)} \cdots a_{n,\pi(n)}$$

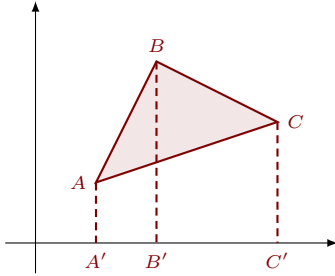
has a factor of 0 if $\pi \neq 123 \cdots n$.

Consequently,

$$\det(I_n) = 1 \text{ and } \det(0_n) = 0.$$

Furthermore, by definition, if a square matrix has a row (or column) of 0's, then its determinant is also 0.

Determinants as Areas of Polygons



Let us start by considering the area of triangle in the plane. Let $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$ be vertices of a triangle in $\mathbb{R}^2$. Let A' , B' and C' be the points with coordinates $(x_1, 0)$, $(x_2, 0)$ and $(x_3, 0)$, respectively. Then the area of triangle ABC is calculated as follows:

$$\begin{aligned} \text{Area}(\triangle ABC) &= \text{Area}(\text{trapezoid } A'ABB') + \text{Area}(\text{trapezoid } B'BCC') \\ &\quad - \text{Area}(\text{trapezoid } A'ACC') \end{aligned}$$

Recall that the area of a trapezoid is calculated by multiplying the average of the lengths of the parallel sides with the distance between the parallel sides. Thus,

$$\begin{aligned} \text{Area}(\triangle ABC) &= \frac{1}{2}(y_1 + y_2)(x_2 - x_1) + \frac{1}{2}(y_2 + y_3)(x_3 - x_2) \\ &\quad - \frac{1}{2}(y_1 + y_3)(x_3 - x_1) \\ &= \frac{1}{2}[x_2y_1 - x_1y_2 + x_3y_2 - x_2y_3 - x_3y_1 + x_1y_3]. \end{aligned}$$

Consider the matrix $A = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$. One checks that the determinant of A equals

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = x_2y_1 - x_1y_2 + x_3y_2 - x_2y_3 - x_3y_1 + x_1y_3.$$

We note that the determinant of A maybe positive or negative depending on the location of the points A , B and C . Thus, we have the following result.

Theorem 114

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be noncollinear points in $\mathbb{R}^2$. The area of $\triangle ABC$ is equal to $\frac{1}{2} |\det(M)|$, where

$$M = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}.$$

Example 115. Calculate the area of the triangle with vertices at $(1, 2)$, $(5, 4)$ and $(3, 6)$.

Solution. By Theorem 114, the area of the triangle is $A = \frac{1}{2} |\det(M)|$, where

$$\det(M) = \begin{vmatrix} 1 & 2 & 1 \\ 5 & 4 & 1 \\ 3 & 6 & 1 \end{vmatrix} = (4 + 6 + 30) - (12 + 6 + 10) = 12.$$

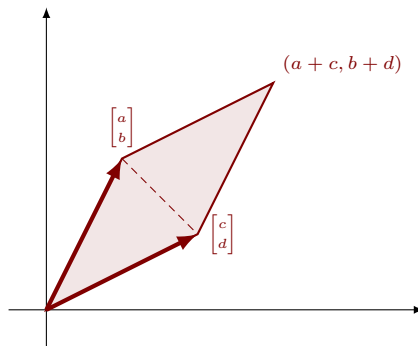
Therefore, $A = \frac{1}{2} \cdot |12| = 6$.



Now, consider the nonzero vectors

$$\vec{v}_1 = \begin{bmatrix} a \\ b \end{bmatrix} \text{ and } \vec{v}_2 = \begin{bmatrix} c \\ d \end{bmatrix}$$

in $\mathbb{R}^2$. These vectors determine a parallelogram with vertices at $(0, 0)$, (a, b) , $(a + c, b + d)$, and (c, d) . Note that the area of this parallelogram is twice the area of the triangle with vertices at $(0, 0)$, (a, b) , and (c, d) .



The area A of the parallelogram is $A = 2 \cdot \frac{1}{2} |\det(M)| = |\det(M)|$, where

$$M = \begin{bmatrix} 0 & 0 & 1 \\ a & b & 1 \\ c & d & 1 \end{bmatrix}.$$

Calculating the determinant of M yields

$$\det(M) = \begin{vmatrix} 0 & 0 & 1 \\ a & b & 1 \\ c & d & 1 \end{vmatrix} = ad - bc = \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

Theorem 116

If $M \in M_2(\mathbb{R})$, then $|\det(M)|$ represents the area of the parallelogram in $\mathbb{R}^2$ determined by the column vectors of M .

In fact, the following is an extension of this theorem for a 3-by-3 matrix. We omit the proof in this text.

Theorem 117

If $M \in M_3(\mathbb{R})$, then $|\det(M)|$ represents the area of the parallelepiped in $\mathbb{R}^3$ determined by the column vectors of M .

Note that the parallelepiped determined by three vectors in $\mathbb{R}^3$ is the 6-sided solid whose opposite faces are congruent parallelograms.

Basic Properties of Determinants

The permutation definition of the determinant is hard to use, especially for large matrices. This lesson presents alternative methods to compute determinants.

Since

Theorem 118

If A is a square matrix, then $\det(A^t) = \det(A)$.

Proof. Let $A = [a_{ij}] \in M_n$. Let $A^t = [b_{ij}]$. Then $b_{ij} = a_{ji}$, for all i and for all j . Thus,

$$\begin{aligned} \det(A^t) &= \sum_{\pi \in S_n} (-1)^\pi b_{1,\pi(1)} b_{2,\pi(2)} \cdots b_{n,\pi(n)} \\ &= \sum_{\pi \in S_n} (-1)^\pi a_{\pi(1),1} a_{\pi(2),2} \cdots a_{\pi(n),n}. \end{aligned}$$

If $\pi \in S_n$, then there exists a unique $\sigma \in S_n$ such that $\sigma(\pi(i)) = i$, for $i = 1, 2, \dots, n$ and $(-1)^\sigma = (-1)^\pi$ (exercise). Thus,

$$(-1)^\pi a_{\pi(1),1} a_{\pi(2),2} \cdots a_{\pi(n),n} = (-1)^\sigma a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}.$$

This proves that $\det(A^t) = \det A$. ▢

Intuitively, every term of $\det(A^t)$ is a signed product of entries of A coming from each column, each appearing in exactly one row. This signed product is also a term in the expansion of $\det(A)$.

We now discuss the effects of elementary row operations on a matrix to its determinant. In the following set of theorems, Theorem 118 guarantees that we may change “rows” into “columns” and still obtain true statements.

Theorem 119

If a matrix B is obtained from a square matrix A by interchanging two distinct rows. Then

$$\det B = -\det A.$$

Proof. Let $A = [a_{ij}] \in M_n$ be given. Fix integers r and s such that $1 \leq r < s \leq n$. Let $B = [b_{ij}] \in M_n$, where

$$b_{ij} = \begin{cases} a_{ij} & \text{if } i \neq r, s; \\ a_{sj} & \text{if } i = r; \\ a_{rj} & \text{if } i = s. \end{cases}$$

In other words, B is the matrix obtained from A by interchanging rows r and s . Thus, for a permutation $\pi \in S_n$,

$$\begin{aligned} b_{1,\pi(1)} \cdots b_{r,\pi(r)} \cdots b_{s,\pi(s)} \cdots b_{n,\pi(n)} &= a_{1,\pi(1)} \cdots a_{s,\pi(r)} \cdots a_{r,\pi(s)} \cdots a_{n,\pi(n)} \\ &= a_{1,\sigma(1)} \cdots a_{r,\sigma(r)} \cdots a_{s,\sigma(s)} \cdots a_{n,\sigma(n)}, \end{aligned}$$

where σ is a permutation such that $\sigma(i) = \pi(i)$, if $i \neq r, s$, $\sigma(r) = \pi(s)$ and $\sigma(s) = \pi(r)$. Thus, $(-1)^\sigma = -(-1)^\pi$. Every term of $\det(B)$ is a negative of a term of $\det(A)$. Therefore, $\det(B) = -\det(A)$. ▣

Example 120. Evaluate $\begin{vmatrix} 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{vmatrix}$.

Solution.

$$\begin{aligned} \begin{vmatrix} 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{vmatrix} &= - \begin{vmatrix} 0 & 2 & 5 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 3 & 0 \end{vmatrix} && \text{(interchange columns 2 and 4)} \\ &= \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 3 & 0 \end{vmatrix} && \text{(interchange rows 1 and 2)} \\ &= - \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{vmatrix} && \text{(interchange rows 3 and 4)} \\ &= -(1 \cdot 2 \cdot 3 \cdot 4) && \text{(determinant of an upper triangular)} \\ &= -24 \end{aligned}$$

Consider a square matrix A with two identical rows. If we interchange these rows, we obtain a matrix B with $\det B = -\det A$. But, since we interchange two equal rows, $\det B = \det A$. Thus, it follows that

$$\det A = -\det A \implies 2\det A = 0 \implies \det A = 0.$$

We therefore have the following consequence.

Corollary 121

If a square matrix A has two equal rows (or columns), then $\det A = 0$.

We next consider the effect on the determinant of a matrix if a row or a column is multiplied by a constant.

Theorem 122

If a matrix B is obtained from a square matrix A by multiplying a row of A by a scalar c , then

$$\det B = c \cdot \det A.$$

Proof. Let $A = [a_{ij}] \in M_n$ and let $B = [b_{ij}]$ be the matrix obtained from A by multiplying row r of A by a scalar c . Then,

$$b_{ij} = \begin{cases} a_{ij} & \text{if } i \neq r; \\ ca_{ij} & \text{if } i = r. \end{cases}$$

Hence,

$$\begin{aligned} \det B &= \sum_{\pi \in S_n} (-1)^\pi b_{1,\pi(1)} \cdots b_{r,\pi(r)} \cdots b_{n,\pi(n)} \\ &= \sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} \cdots ca_{r,\pi(r)} \cdots a_{n,\pi(n)} \\ &= c \sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} \cdots a_{r,\pi(r)} \cdots a_{n,\pi(n)} \\ &= c \cdot \det A. \end{aligned}$$



Example 123. Let $A = \begin{bmatrix} 1 & -2 & 1 \\ 4 & 1 & 9 \\ -2 & 4 & -2 \end{bmatrix}$. Calculate $\det(A)$.

Solution. Theorem [122](#) and Corollary [121](#) guarantee that

$$\begin{vmatrix} 1 & -2 & 1 \\ 4 & 1 & 9 \\ -2 & 4 & -2 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 \\ 4 & 1 & 9 \\ -2(1) & -2(-2) & -2(1) \end{vmatrix} = -2 \begin{vmatrix} 1 & -2 & 1 \\ 4 & 1 & 9 \\ 1 & -2 & 1 \end{vmatrix} = 0.$$

We finally consider the effect on the determinant of a square matrix if one of its rows is replaced by the sum of the row and a multiple of another row.

Theorem 124

If B is a matrix obtained from a square matrix A by replacing a row of A by the sum of the row and a multiple of other row. Then

$$\det B = \det A.$$

Proof. Let $A = [a_{ij}] \in M_n$, $n > 1$ and $B = [b_{ij}]$ be the matrix obtained from A by replacing row r of A by the sum of row r and c times row s , where $1 \leq r < s \leq n$. It follows that $b_{ij} = a_{ij}$, if $i \neq r$ and $b_{rj} = a_{rj} + ca_{sj}$, $j = 1, 2, \dots, n$. Thus,

$$\begin{aligned} \det B &= \sum_{\pi \in S_n} (-1)^\pi b_{1,\pi(1)} \cdots b_{r,\pi(r)} \cdots b_{n,\pi(n)} \\ &= \sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} \cdots (a_{r,\pi(r)} + ca_{s,\pi(r)}) \cdots a_{s,\pi(s)} \cdots b_{n,\pi(n)} \end{aligned}$$

The sum on the right-hand side can then be expanded to

$$\begin{aligned} \det B &= \sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} \cdots a_{r,\pi(r)} \cdots a_{n,\pi(n)} \\ &\quad + c \sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} \cdots a_{s,\pi(r)} \cdots a_{s,\pi(s)} \cdots a_{n,\pi(n)} \end{aligned}$$

Now, the first term

$$\sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} \cdots a_{r,\pi(r)} \cdots a_{n,\pi(n)}$$

is $\det A$ and the sum

$$\sum_{\pi \in S_n} (-1)^\pi a_{1,\pi(1)} \cdots a_{s,\pi(r)} \cdots a_{s,\pi(s)} \cdots a_{n,\pi(n)}$$

is the determinant of a matrix, where row r and row s are equal, hence equal to 0 by Corollary [121](#). Therefore,

$$\det B = \det A + c \cdot 0 = \det A.$$



To summarize, if A is a square matrix and a matrix B is obtained from A by performing a single elementary row operation on A , then the determinant of B is related to the determinant of A as in the following table.

row / column operation	$\det B$
E_{rs}	$-\det A$
$E_r(c)$	$c \det A$
$E_{rs}(c)$	$\det A$

Example 125. Evaluate the *Vandermonde* determinant

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$$

Solution.

$$\begin{aligned} \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} &= \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} \\ &= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 1 & c+a \end{vmatrix} \\ &= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & c-b \end{vmatrix} \\ &= (b-a)(c-a)(c-b). \end{aligned}$$



One common way to evaluate determinants is to use row/column operations in order to reduce a matrix in a triangular form. We illustrate the method in the following example.

Example 126. Evaluate the determinant

$$\begin{vmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 1 & 2 & 3 & 0 \\ 5 & 3 & 2 & -1 \end{vmatrix}.$$

Solution.

$$\begin{vmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 1 & 2 & 3 & 0 \\ 5 & 3 & 2 & -1 \end{vmatrix}$$
$$= \begin{vmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 3 & 1 & -3 \\ 0 & 8 & -8 & -16 \end{vmatrix}$$
$$= 4 \cdot \begin{vmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 3 & 1 & -3 \\ 0 & 8 & -8 & -16 \end{vmatrix}$$
$$= 4 \cdot \begin{vmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & -8 \end{vmatrix}$$
$$= 4 \cdot 1 \cdot 1 \cdot 4 \cdot (-8)$$
$$= -128$$

operations used

$[E_{21}(-2), E_{31}(-1), E_{41}(-5)]$

$[E_2(1/4)]$

$[E_{32}(-3), E_{42}(-8)]$

upper triangular



Recall that an elementary matrix is a matrix which can be obtained by performing a single elementary row operation on the identity matrix. Since the determinant of the identity matrix is 1, we can directly observe the following using the results above.

Lemma 127

Let E be an elementary matrix.

1. If $E = E_{rs}$, then $\det E = -1$.
2. If $E = E_r(c)$ ($c \neq 0$), then $\det E = c$.
3. If $E = E_{rs}(c)$, then $\det E = 1$.

Consequently, $\det E \neq 0$.

Lemma [127](#) now imply the following.

Theorem 128

Let $A \in M_n$ and let $E \in M_n$ be an elementary matrix. Then

$$\det(EA) = \det E \cdot \det A$$

It now follows that if $E_1, E_2, \dots, E_k$ are elementary matrices and $B = E_1 E_2 \cdots E_k A$, then

$$\begin{aligned} \det B &= \det[E_1(E_2 \cdots E_k A)] \\ &= \det E_1 \cdot \det[E_2(E_3 \cdots E_k A)] \\ &\vdots \\ &= \det E_1 \cdot \det E_2 \cdots \det E_k \cdot \det A. \end{aligned}$$

Assume now that A is a nonsingular matrix. Theorem [128](#) guarantees that $A = E_1 E_2 \cdots E_k$, where E_i is an elementary matrix $i = 1, 2, \dots, k$. Since the determinant of an elementary matrix is not zero, it follows that

$$\det A = \det E_1 E_2 \cdots E_k = \det E_1 \cdot \det E_2 \cdots \det E_k \neq 0.$$

Conversely, assume that $\det A \neq 0$. We aim to prove that A is nonsingular. For the sake of contradiction, suppose that A is singular. Then $B = \text{RREF}(A)$ has a row of zeroes. Thus, $\det B = 0$. Since A is row equivalent to B , there exist elementary matrices $E_1, E_2, \dots, E_k$

such that $A = E_1, E_2, \dots, E_k B$. Hence,

$$\det A = \det E_1 \cdot \det E_2 \cdot \dots \cdot \det E_k \cdot \det B = 0,$$

which is a contradiction. We therefore have the following useful result.

Theorem 129

A square matrix A is nonsingular if and only if $\det A \neq 0$.

After all, we have learned that determinant is a good test for nonsingularity. It determines whether a given square matrix is nonsingular or not, hence the name.

Theorem 130

If A and B are n -by- n matrices, then

$$\det(AB) = \det A \cdot \det B.$$

Proof. We prove the result in cases.

(Case 1) A is singular.

If B is singular, then there exists a nonzero $\vec{x} \in \mathbb{R}^n$ such that $B\vec{x} = \vec{0}$ implying that $(AB)\vec{x} = \vec{0}$. Thus, AB is singular.

Now, suppose B is nonsingular. Since A is singular, choose a nonzero $\vec{y} \in \mathbb{R}^n$ such that $A\vec{y} = \vec{0}$. Then choose $\vec{x} \in \mathbb{R}^n$ such that $B\vec{x} = \vec{y}$. It follows that $(AB)\vec{x} = A\vec{y} = \vec{0}$, proving that AB is singular.

Therefore, by Theorem [129](#),

$$\det(AB) = 0 = 0 \cdot \det B = \det A \cdot \det B.$$

(Case 2) A is nonsingular.

If A is nonsingular, write $A = E_1 E_2 \cdots E_k$ as a product of elementary matrices. Then

$$\begin{aligned} \det(AB) &= \det E_1 \det E_2 \cdots \det E_k \det B \\ &= \det(E_1 E_2 \cdots E_k) \det B \\ &= \det A \cdot \det B. \end{aligned}$$

**Corollary 131**

If A and B are n -by- n matrices, then

$$\det(AB) = \det(BA).$$

Proof. $\det(AB) = \det A \cdot \det B = \det B \cdot \det A = \det(BA).$



Notice that Theorem [130](#) may be extended inductively to products of several matrices.

Corollary 132

If $A_1, A_2, \dots, A_k$ are n -by- n matrices, then

$$\det(A_1 A_2 \cdots A_k) = \det A_1 \det A_2 \cdots \det A_k.$$

Finally, if A is nonsingular, then $AA^{-1} = I_n$. Taking determinants of both sides yields $\det A \cdot \det(A^{-1}) = 1$. Since $\det A \neq 0$, the following result follow.

Theorem 133

If $A \in M_n$ is nonsingular, then

$$\det(A^{-1}) = \frac{1}{\det A}.$$

Lesson 2.4: Cofactor Expansion

Learning Outcomes: At the end of the lesson, the students are able to:

1. compute cofactors of entries in a matrix.
2. evaluate determinants using cofactor expansion.
3. solve linear systems using the Cramer's rule.

Cofactors

Consider the expansion of the determinant of a 3-by-3 matrix $[a_{ij}]$.

$$\begin{aligned}
 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\
 &\quad - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} \\
 &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{23}a_{31} - a_{21}a_{33}) \\
 &\quad + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\
 &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) \\
 &\quad + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\
 &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.
 \end{aligned}$$

Observe that we were able to express a 3-by-3 determinant as linear combinations of 2-by-2 determinants. In this section, we generalize this observation to an n -by- n determinant, where $n > 1$. We express the determinant of an n -by- n matrix as a linear combination of $(n - 1)$ -by- $(n - 1)$ determinants. This method of computing determinants is called *cofactor expansion*. Thus, computing determinants of an arbitrary size may be reduced to determinants of size 2-by-2 by repeatedly applying cofactor expansion. This method is useful when we are computing determinants of size n -by- n , where $n > 3$.

If $A = [a_{ij}]$ is an n -by- n matrix, then for each (i, j) , we may write

$$\begin{aligned}
 \det(A) &= (\text{terms containing } a_{ij}) + (\text{terms not containing } a_{ij}) \\
 &= a_{ij} \cdot A_{ij} + (\text{terms not containing } a_{ij}),
 \end{aligned}$$

where A_{ij} is some scalar. The factor A_{ij} is called the **cofactor** of the entry a_{ij} .

In symbols, we may write

$$\det(A) = a_{ij}A_{ij} + \sum_{s \neq j} a_{is}A_{is}.$$

Therefore, we have the following result.

Theorem 134: Cofactor Expansion

Let $A = [a_{ij}]$ be an n -by- n matrix. Then for each $i = 1, 2, \dots, n$,

$$\det(A) = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}.$$

Also, for each $j = 1, 2, \dots, n$,

$$\det(A) = a_{1j}A_{1j} + a_{2j}A_{2j} + \dots + a_{nj}A_{nj}.$$

Computing Cofactors

Let $A = [a_{ij}] \in M_n$ be given. We first calculate the cofactor A_{11} of the $(1, 1)$ -entry. Let $(i, j) = (1, 1)$.

Let $S = \{\pi \in S_n : \pi(1) = 1\}$. The set S consists of permutations fixing entry 1. In other words, for every $\pi \in S$,

$$(-1)^\pi a_{1,\pi(1)} a_{2,\pi(2)} \cdots a_{n,\pi(n)} = (-1)^\pi a_{11} a_{2,\pi(2)} \cdots a_{n,\pi(n)}$$

is a term of $\det(A)$ that contains a_{11} as a factor. It follows that

$$\sum_{\pi \in S} (-1)^\pi a_{1,\pi(1)} a_{2,\pi(2)} \cdots a_{n,\pi(n)}$$

consists of all terms in $\det(A)$ that has a_{11} as a factor. Thus,

$$\begin{aligned}
\det(A) &= \sum_{\pi \in S_n} (-1)^\pi a_{1\pi(1)} \cdots a_{n\pi(n)} \\
&= \sum_{\pi \in S} (-1)^\pi a_{1\pi(1)} \cdots a_{n\pi(n)} + \sum_{\pi \in S_n \setminus S} (-1)^\pi a_{1\pi(1)} \cdots a_{n\pi(n)} \\
&= \sum_{\pi \in S} (-1)^\pi a_{11} \cdots a_{n\pi(n)} + \sum_{\pi \in S_n \setminus S} (-1)^\pi a_{1\pi(1)} \cdots a_{n\pi(n)} \\
&= a_{11} \sum_{\pi \in S} a_{2\pi(2)} \cdots a_{n\pi(n)} + \sum_{\pi \in S_n \setminus S} (-1)^\pi a_{1\pi(1)} \cdots a_{n\pi(n)}.
\end{aligned}$$

Consequently, the cofactor of a_{11} is

$$A_{11} = \sum_{\pi \in S} a_{2\pi(2)} \cdots a_{n\pi(n)}.$$

Since $\pi \in S$ defines a permutation on $\{2, 3, \dots, n\}$, then A_{11} is the determinant

$$\begin{vmatrix}
a_{22} & a_{23} & \cdots & a_{2n} \\
a_{32} & a_{33} & \cdots & a_{3n} \\
\vdots & \vdots & & \vdots \\
a_{n2} & a_{n3} & \cdots & a_{nn}
\end{vmatrix},$$

which is the determinant of $(n-1)$ -by- $(n-1)$ matrix obtained by deleting the first row and first column of A .

Let $(i, j) \neq (1, 1)$. We next consider the matrix B obtained from A by performing the following sequence of operations:

- For each $k = 1, 2, \dots, j$, interchange the position of column k and column $k-1$.

This moves the column j of A to the first column position, while maintaining the relative positions of the other columns. In effect, the entry a_{ij} is moved to the first column maintaining its row number. To the resulting matrix, we perform the following:

- For $\ell = 1, 2, \dots, i$, interchange the position of row ℓ and row $\ell-1$.

Thus puts the entry a_{ij} in the $(1, 1)$ -position while the relative positions of other rows and columns (except for row i and column j) are respected. Let B be the resulting matrix.

We can visualize B as follows.

$$B = \left[\begin{array}{c|cccccc} a_{ij} & a_{i,1} & \cdots & a_{i,j-1} & a_{i,j+1} & \cdots & a_{i,n} \\ \hline a_{1,j} & & & & & & \\ \vdots & & & & & & \\ a_{i-1,j} & & & & & & \\ a_{i+1,j} & & & & M & & \\ \vdots & & & & & & \\ a_{n,1} & & & & & & \end{array} \right],$$

where M is the $(n-1)$ -by- $(n-1)$ matrix obtained by deleting row i and column j of A . Since B is obtained by performing $(i-1)$ row swaps and $(j-1)$ column swaps, it follows that

$$\det B = (-1)^{(i-1)+(j-1)} \det A = (-1)^{i+j} \det A.$$

Following the argument on the cofactor of the $(1,1)$ entry of a matrix, the cofactor of a_{ij} as an entry of matrix B is $\det M$. Hence, the cofactor of a_{ij} as an entry of the matrix A is

$$A_{ij} = (-1)^{i+j} \det M,$$

where M is the matrix obtained by deleting the row i and column j of A . The determinant of the matrix M is called the **minor** of the entry a_{ij} of A . We summarize this in the following theorem.

Theorem 135: Formula for the Cofactor of a_{ij}

If $A = [a_{ij}]$ is an n -by- n matrix, then for all i and for all j , the cofactor of the entry a_{ij} is given by

$$A_{ij} = (-1)^{i+j} \det M_{ij},$$

where M_{ij} is the $(n-1)$ -by- $(n-1)$ matrix obtained by deleting the row i and the column j of A .

Example 136. Using cofactor expansion, evaluate the following determinants.

$$1. \begin{vmatrix} 2 & 3 & -1 \\ 0 & 1 & -1 \\ 3 & 0 & 2 \end{vmatrix}$$

$$2. \begin{vmatrix} 9 & 1 & 9 & 9 & 9 \\ 9 & 0 & 9 & 9 & 2 \\ 4 & 0 & 0 & 5 & 0 \\ 9 & 0 & 3 & 9 & 0 \\ 6 & 0 & 0 & 7 & 0 \end{vmatrix}$$

Solutions.

1. Using cofactor expansion along the second row, we see that

$$\begin{vmatrix} 2 & 3 & -1 \\ \mathbf{0} & \mathbf{1} & \mathbf{-1} \\ 3 & 0 & 2 \end{vmatrix} = \mathbf{0} \cdot (-1)^{2+1} \begin{vmatrix} 3 & -1 \\ 0 & 2 \end{vmatrix} + \mathbf{1} \cdot (-1)^{2+2} \begin{vmatrix} 2 & -1 \\ 3 & 2 \end{vmatrix} + \\ + \mathbf{(-1)} \cdot (-1)^{2+3} \begin{vmatrix} 2 & 3 \\ 3 & 0 \end{vmatrix}$$

$$= 0 + 7 + -9 = -2$$

2. The idea in the cofactor expansion is to choose the row (or column) which contains the most number of zeroes so that the computation for the determinant is less complicated. To proceed, we first expand along the second column.

$$\begin{vmatrix} 9 & \mathbf{1} & 9 & 9 & 9 \\ 9 & \mathbf{0} & 9 & 9 & 2 \\ 4 & \mathbf{0} & 0 & 5 & 0 \\ 9 & \mathbf{0} & 3 & 9 & 0 \\ 6 & \mathbf{0} & 0 & 7 & 0 \end{vmatrix} = \mathbf{1} \cdot (-1)^{1+2} \begin{vmatrix} 9 & 9 & 9 & 2 \\ 4 & 0 & 5 & 0 \\ 9 & 3 & 9 & 0 \\ 6 & 0 & 7 & 0 \end{vmatrix} = - \begin{vmatrix} 9 & 9 & 9 & \mathbf{2} \\ 4 & 0 & 5 & \mathbf{0} \\ 9 & 3 & 9 & \mathbf{0} \\ 6 & 0 & 7 & \mathbf{0} \end{vmatrix}$$

We next expand along the fourth column of the resulting determinant.

$$\begin{vmatrix} 9 & 9 & 9 & \mathbf{2} \\ 4 & 0 & 5 & \mathbf{0} \\ 9 & 3 & 9 & \mathbf{0} \\ 6 & 0 & 7 & \mathbf{0} \end{vmatrix} = \mathbf{2} \cdot (-1)^{1+4} \begin{vmatrix} 4 & 0 & 5 \\ 9 & 3 & 9 \\ 6 & 0 & 7 \end{vmatrix} = -2 \begin{vmatrix} 4 & \mathbf{0} & 5 \\ 9 & \mathbf{3} & 9 \\ 6 & \mathbf{0} & 7 \end{vmatrix}$$

We next expand along the second column of the resulting determinant.

$$\begin{vmatrix} 4 & \mathbf{0} & 5 \\ 9 & \mathbf{3} & 9 \\ 6 & \mathbf{0} & 7 \end{vmatrix} = \mathbf{3} \cdot (-1)^{2+2} \begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix} = 3(28 - 30) = -6.$$

Using *back substitution*, we see that

$$\begin{vmatrix} 9 & 1 & 9 & 9 & 9 \\ 9 & 0 & 9 & 9 & 2 \\ 4 & 0 & 0 & 5 & 0 \\ 9 & 0 & 3 & 9 & 0 \\ 6 & 0 & 0 & 7 & 0 \end{vmatrix} = -[-2(-6)] = -12.$$

The previous set of examples illustrates how we can apply cofactor expansion repeatedly reducing the problem of computing the determinant of a *large* matrix into sums of determinants of *smaller* matrices.

Example 137. Given that $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 5$, calculate the determinant

$$\begin{vmatrix} a & 0 & 0 & b \\ 0 & a & b & 0 \\ 0 & c & d & 0 \\ c & 0 & 0 & d \end{vmatrix}.$$

Solution. Using cofactor expansion along the first row, we see that

$$\begin{aligned} \begin{vmatrix} a & 0 & 0 & b \\ 0 & a & b & 0 \\ 0 & c & d & 0 \\ c & 0 & 0 & d \end{vmatrix} &= a \cdot (-1)^{1+1} \begin{vmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & d \end{vmatrix} + b \cdot (-1)^{1+4} \begin{vmatrix} 0 & a & b \\ 0 & c & d \\ c & 0 & 0 \end{vmatrix} \\ &= a \begin{vmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & d \end{vmatrix} - b \begin{vmatrix} 0 & a & b \\ 0 & c & d \\ c & 0 & 0 \end{vmatrix} \end{aligned}$$

Cofactor expansion along the third column yields

$$\begin{vmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & d \end{vmatrix} = d \cdot (-1)^{3+3} \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 5d.$$

On the other hand, cofactor expansion along the first column yields

$$\begin{vmatrix} 0 & a & b \\ 0 & c & d \\ c & 0 & 0 \end{vmatrix} = c \cdot (-1)^{3+1} \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 5c.$$

Substituting these values to the original equation yields

$$\begin{vmatrix} a & 0 & 0 & b \\ 0 & a & b & 0 \\ 0 & c & d & 0 \\ c & 0 & 0 & d \end{vmatrix} = a(5d) - b(5c) = 5(ad - bc) = 5 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 5 \cdot 5 = 25.$$

Computing Inverse of a Matrix Using Determinants

We now discuss a method that computes the inverse of a nonsingular matrices using cofactors.

Recall that if $A = [a_{ij}]$ is an n -by- n matrix and A_{ij} is the cofactor of the entry a_{ij} for all i and for all j , then for each i ,

$$\det A = \sum_{j=1}^n a_{ij} A_{ij} = a_{i1} A_{i1} + a_{i2} A_{i2} + \cdots + a_{in} A_{in},$$

and for each j ,

$$\det A = \sum_{i=1}^n a_{ij} A_{ij} = a_{1j} A_{1j} + a_{2j} A_{2j} + \cdots + a_{nj} A_{nj}.$$

Fix an $i \in \{1, 2, \dots, n\}$ and let $r \neq i$. Let $A = [a_{ij}] \in M_{m,n}$ such that $a_{rj} = a_{ij}$, for $j = 1, 2, \dots, n$, that is, row r of A is equal to its row i . Since two rows of A are identical,

it follows that $\det A = 0$. Using cofactor expansion along row r , we see that

$$\begin{aligned}\det A &= a_{r1}A_{r1} + a_{r2}A_{r2} + \cdots + a_{rn}A_{rn} \\ &= a_{i1}A_{r1} + a_{i2}A_{r2} + \cdots + a_{in}A_{rn}.\end{aligned}$$

Hence, we have the following observation.

Lemma 138

Let $A = [a_{ij}]$ be an n -by- n matrix. Then, for each $i = 1, 2, \dots, n$,

$$\sum_{j=1}^n a_{ij}A_{rj} = \begin{cases} 0 & \text{if } r \neq i; \\ \det A & \text{if } r = i. \end{cases}$$

Similarly, for each $j = 1, 2, \dots, n$, we have

$$\sum_{i=1}^n a_{ij}A_{ic} = \begin{cases} 0 & \text{if } c \neq j; \\ \det A & \text{if } c = j. \end{cases}$$

Definition 139: Adjugate

If $A = [a_{ij}]$ is an n -by- n matrix and if A_{ij} is the cofactor of the entry a_{ij} , the **adjugate** of A is the matrix

$$\operatorname{adj} A = [A_{ij}]^t.$$

Note that if $A = [a_{ij}] \in M_n$, then

$$\operatorname{adj}(A) = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}.$$

Example 140. We compute $\text{adj}(A)$, where $A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & -2 & 3 \end{bmatrix}$. We get the cofactors of each entry. For example, the cofactor of the $(1, 1)$ - entry is $\begin{vmatrix} 1 & 0 \\ -2 & 3 \end{vmatrix} = 3$. The cofactor of the $(1, 2)$ - entry is $-\begin{vmatrix} 1 & 0 \\ 0 & 3 \end{vmatrix} = -3$. And so on. We summarize the cofactors in the following table.

entry	cofactor	entry	cofactor	entry	cofactor
$(1, 1)$	$\begin{vmatrix} 1 & 0 \\ -2 & 3 \end{vmatrix} = 3$	$(2, 1)$	$-\begin{vmatrix} 0 & -1 \\ -2 & 3 \end{vmatrix} = 2$	$(3, 1)$	$\begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = 1$
$(1, 2)$	$-\begin{vmatrix} 1 & 0 \\ 0 & 3 \end{vmatrix} = -3$	$(2, 2)$	$\begin{vmatrix} 2 & -1 \\ 0 & 3 \end{vmatrix} = 6$	$(3, 2)$	$-\begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} = -1$
$(1, 3)$	$\begin{vmatrix} 1 & 1 \\ 0 & -2 \end{vmatrix} = -2$	$(2, 3)$	$-\begin{vmatrix} 2 & 0 \\ 0 & -2 \end{vmatrix} = 4$	$(3, 3)$	$\begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2$

Therefore, $\text{adj } A = \begin{bmatrix} 3 & 2 & 1 \\ -3 & 6 & -1 \\ -2 & 4 & 2 \end{bmatrix}$. Observe that if we multiply A to $\text{adj } A$ we get

$$A(\text{adj } A) = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & -2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ -3 & 6 & -1 \\ -2 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix} = 8I_3,$$

and $\det A = 8$.

This observation leads to the following general result.

Theorem 141

Let A be an n -by- n matrix. Then

$$A(\operatorname{adj} A) = (\det A)I_n = (\operatorname{adj} A)A.$$

Proof. Let $A = [a_{ij}] \in M_n$ and let $\operatorname{adj} A = [A_{ij}]^T = [A_{ji}]$, where A_{ij} is the cofactor of a_{ij} . If $A(\operatorname{adj} A) = [c_{ij}]$, then for all i and for all j , we have by Lemma 138,

$$c_{ij} = \sum_{k=1}^n a_{ik}A_{jk} = \begin{cases} \det A & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

This implies that $A(\operatorname{adj} A)$ is a diagonal matrix with diagonal entries all $\det A$, so $A(\operatorname{adj} A) = (\det A)I_n$. A similar proof can be done for the equality $(\operatorname{adj} A)A = (\det A)I_n$. $\square$

A nice consequence of this theorem may be derived if we assume that A is nonsingular. If A is nonsingular, then $\det A \neq 0$. If we set $B = \frac{1}{\det A}(\operatorname{adj} A)$, Theorem 141 implies that

$$AB = A \left[\frac{1}{\det A}(\operatorname{adj} A) \right] = \frac{1}{\det A}[A(\operatorname{adj} A)] = \frac{1}{\det A}(\det A)I_n = I_n.$$

Similarly, $BA = I_n$. Consequently, $B = A^{-1}$.

Corollary 142: Explicit Formula for A^{-1}

If A is nonsingular, then

$$A^{-1} = \frac{1}{\det A}(\operatorname{adj} A).$$

Example 143. In the previous example, we see that if

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & -2 & 3 \end{bmatrix},$$

then $\text{adj } A = \begin{bmatrix} 3 & 2 & 1 \\ -3 & 6 & -1 \\ -2 & 4 & 2 \end{bmatrix}$. By direct computation, we can see that

$$\begin{vmatrix} 2 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & -2 & 3 \end{vmatrix} = 8 \text{ implying that } A^{-1} = \frac{1}{8} \begin{bmatrix} 3 & 2 & 1 \\ -3 & 6 & -1 \\ -2 & 4 & 2 \end{bmatrix}.$$

Cramer's Rule

We now illustrate how we may use determinants to solve linear systems of the form $A\vec{x} = \vec{b}$, where A is nonsingular.

We again consider the n -by- n linear system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n, \end{aligned}$$

which in matrix form can be expressed as $A\vec{x} = \vec{b}$, where $A = [a_{ij}] \in M_n$, $\vec{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^t$ and $\vec{b} = \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}^t$. Recall that if the matrix A is nonsingular, then the linear system $A\vec{x} = \vec{b}$ has a unique solution $\vec{x} = A^{-1}\vec{b}$. By Corollary

[142](#),

$$A^{-1} = \frac{1}{\det A}(\text{adj } A) \text{ implying that } \vec{x} = \frac{1}{\det A}(\text{adj } A)\vec{b}.$$

So,

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{\det A} \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

By direct computation, we see that for $j = 1, 2, \dots, n$, we have

$$x_j = \frac{b_1 A_{1j} + b_2 A_{2j} + \cdots + b_n A_{nj}}{\det A}. \quad (2.8)$$

Now, for $j = 1, 2, \dots, n$, let

$$A_j = \begin{bmatrix} a_{11} & \cdots & a_{1,j-1} & b_1 & a_{1,j+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2,j-1} & b_2 & a_{2,j+1} & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \cdots & a_{n,j-1} & b_n & a_{n,j+1} & \cdots & a_{nn} \end{bmatrix},$$

that is, A_j is a matrix obtained from A by replacing the j th column by $\vec{b}$. Getting the determinant of D_j by cofactor expansion along the j th column, we see that

$$\det A_j = b_1 A_{1j} + b_2 A_{2j} + \cdots + b_n A_{nj}.$$

Equation 2.8 now implies that

$$x_j = \frac{\det A_j}{\det A}.$$

We now have established the **Cramer's Rule**.

Theorem 144: Cramer's Rule

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

If A is nonsingular and $A\vec{x} = \vec{b}$, then for $j = 1, 2, \dots, n$,

$$x_j = \frac{\det A_j}{\det A},$$

where

$$A_j = \begin{bmatrix} a_{11} & \cdots & a_{1,j-1} & b_1 & a_{1,j+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2,j-1} & b_2 & a_{2,j+1} & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \cdots & a_{n,j-1} & b_n & a_{n,j+1} & \cdots & a_{nn} \end{bmatrix}$$

is the matrix obtained from A by replacing the j th column by $\vec{b}$.

Example 145. Solve the linear system

$$\begin{cases} 2x + 3y = 2 \\ x + 2y = 0. \end{cases}$$

Solution. Let $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. Then the given linear system is equivalent to $A\vec{x} = \vec{b}$. Since $\det A = (2)(2) - (3)(1) = 1 \neq 0$, the Cramers rule implies that

$$x = \frac{\begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix}}{1} = 4 \text{ and } y = \frac{\begin{vmatrix} 2 & 2 \\ 1 & 0 \end{vmatrix}}{1} = -2.$$

Example 146. Find the solution set of the linear system

$$\begin{cases} 2x + y + z = 6 \\ 3x + 2y - 2z = -2. \\ x + y + 2z = -4 \end{cases}$$

Solution. If we let $A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & -2 \\ 1 & 1 & 2 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 6 \\ -2 \\ -4 \end{bmatrix}$, then the given linear system is equivalent to $A\vec{x} = \vec{b}$. Since

$$\det A = \begin{vmatrix} 2 & 1 & 1 \\ 3 & 2 & -2 \\ 1 & 1 & 2 \end{vmatrix} = 5 \neq 0,$$

we can apply the Cramer's rule to get the unique values for x , y and z as follows.

$$x = \frac{\begin{vmatrix} 6 & 1 & 1 \\ -2 & 2 & -2 \\ -4 & 1 & 2 \end{vmatrix}}{5} = \frac{54}{5}, y = \frac{\begin{vmatrix} 2 & 6 & 1 \\ 3 & -2 & -2 \\ 1 & -4 & 2 \end{vmatrix}}{5} = \frac{-82}{5}, \text{ and } z = \frac{\begin{vmatrix} 2 & 1 & 6 \\ 3 & 2 & -2 \\ 1 & 1 & -4 \end{vmatrix}}{5} = \frac{4}{5}.$$

